Q1) Write an SQL query using the WITH clause to create a temporary result set called TopArtists that selects the top 3 most-followed artists from a table SpotifyArtists (columns: artist_id, name, followers), then select all rows from TopArtists.

```sql
WITH TopArtists AS (
    SELECT artist_id,
name, followers
    FROM SpotifyArtists
    ORDER BY followers
DESC
    LIMIT 3
)
SELECT *
FROM TopArtists;
```

Q2) Using a table named FlipkartOrders (columns: order_id, user_id, order_date, total_amount), write a query with a CTE called MonthlyTotals that calculates the total sales amount for each month in 2023, then select the month with the highest total sales from MonthlyTotals.

Hint: Use EXTRACT(MONTH FROM order_date) or DATE_FORMAT(order_date, '%m') depending on your SQL dialect.

```sql
WITH MonthlyTotals AS (
    SELECT
        MONTH(order_date)
AS month,
        SUM(total_amount)
AS total_sales
    FROM FlipkartOrders
    WHERE order_date >=
'2023-01-01'
      AND order_date <
'2024-01-01'
    GROUP BY
MONTH(order_date)
)
SELECT month, total_sales
FROM MonthlyTotals
ORDER BY total_sales DESC
LIMIT 1;
```

Q3) Create a recursive CTE to generate a list of dates for the next 7 days starting from today in a table called CalendarDays (columns: day_date).

Hint: Use UNION ALL inside your CTE and limit recursion to 7 levels.

```
WITH RECURSIVE
CalendarDays AS (
    SELECT CURDATE() AS
day_date

    UNION ALL

    SELECT
DATE_ADD(day_date,
INTERVAL 1 DAY)
    FROM CalendarDays
    WHERE day_date <
DATE_ADD(CURDATE(),
INTERVAL 6 DAY)
)
SELECT day_date
FROM CalendarDays;
```

Q4) Given a table called ZomatoRestaurants (columns: id, name, city, rating), use a CTE to find the average rating for each city, then select all restaurants in cities where the average rating is above 4.0.

```
WITH CityAverage AS (
    SELECT
        city,
        AVG(rating) AS
average_rating
    FROM
ZomatoRestaurants
    GROUP BY city
)
SELECT z.id, z.name,
z.city, z.rating
FROM ZomatoRestaurants z
JOIN CityAverage c
    ON z.city = c.city
WHERE c.average_rating >
4.0;
```

Q5) Use ChatGPT or Copilot to help you write an SQL query with a CTE that, given a table IPLMatches (columns: match_id, team, runs, match_year), calculates the total runs scored by each team in 2023 and lists only teams with more than 2000 runs.

```sql
WITH TeamTotalRuns AS (
    SELECT
        team,
        SUM(runs) AS
total_runs
    FROM IPLMatches
    WHERE match_year =
2023
    GROUP BY team
)
SELECT team, total_runs
FROM TeamTotalRuns
WHERE total_runs > 2000
ORDER BY total_runs DESC;
```